

INDUSTRIAL CROP PRODUCTION II (AGT 214)

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OBJECTIVES

By the end of this unit, you will be able to:

- Identify different types of Industrial tree crops.
- Identify areas of production of various Industrial tree crops.
- Understand the botany of important Industrial tree crops.
- Explain the production techniques of Industrial tree crops in Nigeria.
- Understand the production cycle of major Industrial tree crops in Nigeria.

Tree crops include Cocoa, Rubber, Coffee, Oil Palm, kola, Raffia palm, Cashew, Coconut etc.

DESCRIPTION OF COCOA



Theobroma cacao (cacao tree or cocoa tree) is a small (6–12 m (20–39 ft) tall) evergreen tree in the Malvaceae family.] Its seeds—cocoa beans when dried and fermented—are used to make chocolate liquor, cocoa powder, cocoa butter and chocolate.

HISTORY

The cacao tree was first domesticated at least 5,300 years ago by the Mayo-Chinchipe culture in South America before it was introduced in Mesoamerica.[2] Cacao was consumed by pre-Hispanic cultures in spiritual ceremonies, and its beans were a common currency in Mesoamerica. The cacao tree grows in a limited geographical zone; today, West Africa produces nearly 81% of the world's crop. The three main varieties of cocoa plants are Forastero, Criollo, and Trinitario, with Forastero being the most widely used.

ADAPTATION TO NIGERIA CLIMATIC CONDITIONS

Cocoa adapts to Nigeria's climate through specific farm practices like shade management, rehabilitation of old trees, and adjustments to planting patterns to counter rising temperatures and irregular rainfall. Over 80% of Nigerian cocoa grows in the southwest and southern rainforest belts, requiring careful handling of heat, humidity, and dry spells.

PRODUCING AREAS

Cocoa production is important to the economy of Nigeria. Cocoa is the leading agricultural export of the country and Nigeria is currently the world's fourth largest producer of cocoa, after Ivory Coast, Indonesia and Ghana, and the third largest exporter, after Ivory Coast and Ghana. The crop was a major foreign exchange earner for Nigeria in the 1950s and 1960s and in 1970 the country was the second largest producer in the world but following investments in the oil sector in the 1970s and 1980s, Nigeria's share of world output declined. In 2010, cocoa production accounted for only 0.3% of agricultural GDP. Average cocoa beans production in Nigeria between 2000 and 2010 was 389,272 tonnes per year rising from 170,000 tonnes produced in 1999.

Cocoa production in Nigeria is concentrated primarily in the southern and southwestern rainforest zones, with the top producing states being Ondo, Cross River, and Osun. Major Cocoa-Producing Regions and States

Ondo State: The leading producer in Nigeria, contributing the highest volume of annual output. Key areas include Idanre, Ondo West, Odigbo, Owo, and Akure South local government areas.

Cross River State: The second-largest producer and a major hub in the South-South zone with expanding plantations in Ikom, Boki, Etung, and Obubra.

Osun State: A traditional and high-output southwestern producer recognized for organized farmer cooperatives and well-fermented quality beans.

Oyo State: A long-standing producer across multiple agrarian rural communities in the southwest region.

Ogun State: Contributes steady regional volumes with a geographic advantage for moving produce closer to major port facilities.

Ekiti State: Produces smaller overall quantities but is noted for specialized high-quality, high-yielding cocoa beans.

Edo State & Akwa Ibom State: Notable secondary southern producers with active smallholder farming clusters.

Other States: Minor or emerging production occurs in Delta, Abia, Kwara, Kogi, and Taraba states.

BOTANY OF COCOA

Taxonomy- Scientific Classification

Kingdom: Plantae (plants)

Division: Tracheophyta (vascular plants)

Class: Magnoliopsida (flowering plants / dicotyledons)

Order: Malvales

Family: Malvaceae (sometimes historically placed in Sterculiaceae)

Genus: Theobroma

Species: *Theobroma cacao* L.

MORPHOLOGY/ANATOMY OF COCOA

The morphology of the cocoa plant (*Theobroma cacao*) features an upright main stem, broad alternate leaves, and cauliflorous flowers that develop into large pods. It is a small evergreen tree typically growing 4 to 12 meters tall.

Root System Taproot: Grows straight down deep into the soil to anchor the tree and find water. Lateral roots: Spread out close to the top layer of soil to absorb nutrients

Stem and Architecture Orthotropic shoot (Chupon): The main vertical stem that grows straight up. Plagiotropic branches (Fan branches): Side branches that grow outward in a whorl pattern (called a jorquette), forming the canopy.

Leaves Arrangement: Alternate and simple.

Shape: Oblong to elliptic with a pointed tip and rounded base.

Size: Large, measuring 12 to 60 cm long and 4 to 20 cm wide.

Color: Young leaves (flushes) start as a soft reddish-pink before turning dark, shiny green at maturity.

Flowers Cauliflory: Flowers grow directly on the old wood of the trunk and thick branches rather than at branch tips.

Structure: Very small (about 1 to 2 cm wide), shaped like tiny stars or orchids, and colored white, pink, or yellowish-white.

Pollination: Pollinated by tiny insects called midges rather than bees.

Fruit (Cocoa Pod) Shape: Oval to oblong with 5 to 10 distinct longitudinal ridges or furrows.

Size: 10 to 32 cm long.

Color: Changes from green or white when unripe to bright yellow, orange, red, or purple when mature.

Seeds (Cocoa Beans)Quantity: 20 to 60 seeds packed tightly inside the pod.

Structure: Ovoid to elliptic, enclosed in a sticky, sweet-tart white pulp, containing large fleshy cotyledons

VARIETIES

Traditional Cocoa Varieties

Criollo: Rare, delicate, and produces high-quality aromatic beans, but has low disease resistance and low yield.

Forastero: Robust, highly resilient, and widely grown globally (including the classic West African Amelonado), accounting for the bulk of commercial production.

Trinitario: A natural hybrid of Criollo and Forastero that combines good disease resistance with fine flavor profiles. Improved

Recommended Varieties (CRIN Hybrids)

CRIN TC-1 to TC-8 Series: Released by the Cocoa Research Institute of Nigeria, these improved hybrids bear fruit early (in about 18 to 36 months compared to 5 years for older types), yield 1,500 to 2,300 kg per hectare, and tolerate major pests, drought, and black pod disease.

Synthetic Series (CSS I, CSS II, CSS III): Bred for high heterosis, quick establishment, and specific use in rehabilitating farms impacted by swollen shoot disease.

CULTIVATION PRACTICES

Site Selection

Choosing the right environment is critical for a productive cocoa farm, as cocoa is highly sensitive to its soil and climate.

Soil: Cocoa requires deep, well-drained, and fertile soils rich in organic matter. Loamy soils with good water-holding capacity are ideal. The optimal soil pH is between 6.0 and 7.5. Avoid heavy clay, sandy, or waterlogged soils.

Climate: It thrives in tropical climates with an annual rainfall of 1,250mm to 3,000mm, well-distributed throughout the year. The ideal temperature range is between 21°C and 32°C.

Topography: Flat or gently sloping lands are best to prevent severe soil erosion.

Land Preparation

Proper land preparation ensures the soil is ready to support both the young cocoa seedlings and the necessary shade plants.

Clearing: Clear the land by slashing undergrowth and removing small trees. It is highly recommended to avoid burning the slashed vegetation, as decaying matter adds valuable humus to the soil.

Shade Establishment: Cocoa is a shade-loving plant, especially in its early years. Retain large, high-canopy forest trees to provide natural shade. Alternatively, plant temporary shade crops like plantains, bananas, or cocoyam a few months before transplanting the cocoa.

The Nursery

Because cocoa seeds lose their viability quickly (usually within a week of extraction from the pod), they are typically raised in a nursery before being moved to the main field.

Bagging: Fill perforated black polythene bags (typically 20cm x 12cm) with rich, sieved topsoil.

Sowing: Extract beans from ripe, healthy pods. Remove the sweet mucilage (pulp) by rubbing the beans in sawdust or sand, then plant them horizontally or with the broader end facing downwards, about 2cm to 3cm deep in the soil bags.

Nursery Management

Seedlings spend about 4 to 6 months in the nursery. Proper care during this period dictates the health of the mature tree.

Shade: Construct a nursery shed using palm fronds, bamboo, or shade nets to protect the delicate seedlings from direct sunlight and heavy rainfall. Gradually reduce the shade (hardening off) a month before transplanting.

Watering: Water lightly and regularly (usually early morning or late evening). Keep the soil moist but strictly avoid waterlogging, which causes root rot.

Weeding: Hand-pull weeds from the polybags regularly to prevent competition for nutrients.

Pest and Disease Control: Monitor for insect damage or fungal infections (like Phytophthora) and apply appropriate organic or chemical fungicides/insecticides if necessary.

Planting Date

Nursery Sowing: Seeds are usually sown in the nursery between December and January

Field Planting (Transplanting): Transplanting occurs at the onset of the continuous rainy season, which typically falls between April and June in West Africa, ensuring the young plants have plenty of moisture to establish their root systems.

Lining Out and Pegging

This process ensures plants are arranged in neat rows, which makes future farm operations (weeding, harvesting, and spraying) much easier.

- * Use measuring tapes, ropes, and a builder's square to mark out straight lines across the field.

- * Drive a wooden peg into the ground at every point where a cocoa seedling will be planted.

Spacing

Proper spacing minimizes competition for sunlight, water, and nutrients while maximizing yield per hectare.

- * The standard recommended spacing for cocoa is **3m x 3m** (10ft x 10ft).

- * This spacing provides a plant population of approximately 1,111 cocoa trees per hectare

Holing

Holing prepares the exact spot where the seedling's root ball will sit.

Timing: Dig the holes a few weeks before transplanting to allow the soil to aerate.

Dimensions: Dig holes that are approximately 45cm x 45cm x 45cm

Soil Separation: When digging, keep the fertile topsoil on one side of the hole and the subsoil on the other.

Refilling: Just before planting, mix the topsoil with organic compost or well-rotted manure and use it to partially refill the hole.

Transplanting

Moving the seedling from the nursery to the main field must be done with great care to avoid transplant shock.

Timing: Transplant early in the morning or late in the afternoon on a cloudy or rainy day.

Preparation: Water the seedlings in the nursery heavily the day before moving them so the soil binds together around the roots.

Planting: Use a sharp blade to cut and carefully remove the polythene bag, ensuring the ball of earth around the roots does not break.

Place the seedling into the center of the prepared hole. Ensure the collar of the seedling (where the stem meets the root) is level with the surrounding soil surface.

Fill the remaining space with soil, gently but firmly pressing it down around the base to eliminate air pockets. Apply mulch around the base to retain moisture, ensuring the mulch does not touch the stem directly.

MANAGEMENT

Pruning

Pruning shapes the tree, ensures adequate sunlight penetration, improves air circulation (which reduces humidity-loving fungal diseases), and makes harvesting easier.

Formative Pruning: Done in the early years to encourage the main stem to branch (form a jorquette) at a convenient height, usually between 1.5m and 2m

Sanitary Pruning: The routine removal of dead, diseased, or broken branches, as well as parasitic plants like mistletoe.

Desuckering: The regular and prompt removal of chupons (water suckers) that grow from the base of the trunk. These suckers compete with the main tree for vital nutrients and must be cut off flush with the stem.

Weeding

Weeds compete with cocoa for water, nutrients, and sunlight, especially before the cocoa canopy closes over.

Frequency: Weeding is typically required 3 to 4 times a year during the first three years of establishment. Once the canopy closes, weed growth is naturally suppressed by the shade.

Method: Ring weeding (clearing a circle around the base of the tree) is highly recommended.

Caution: Cocoa has a shallow surface root system (feeder roots). Avoid deep hoeing or aggressive mechanical weeding near the trunk, as this will damage the roots. Slashing with a cutlass or using approved, carefully applied herbicides are the preferred methods.

Mulching

Mulching is essential for young cocoa plants, particularly during the dry season.

Application: Gather dead plantain leaves, slashed weeds, or other dry organic matter and spread them on the soil surface around the base of the tree.

Benefits: It conserves soil moisture, regulates soil temperature, adds organic matter to the soil as it decays, and suppresses weed growth.

Precaution: Ensure the mulch does not directly touch the stem of the cocoa seedling, as this can trap moisture against the bark and encourage collar rot or termite attacks. Keep it a few centimeters away.

Manuring and Fertilizing

Cocoa is a nutrient-demanding crop. High yields extract significant amounts of Potassium, Nitrogen, and Phosphorus from the soil.

Organic Manure: Incorporating well-rotted compost or poultry droppings improves soil structure, water retention, and microbial activity while providing a slow release of nutrients.

Inorganic Fertilizers: Application of NPK fertilizers is standard practice. Depending on soil analysis, formulas like NPK 15:15:15 or specialized cocoa blends (like NPK 0:15:15 if natural nitrogen levels are sufficient) are used.

Timing and Method: Fertilizers are best applied twice a year: once at the onset of the rains and once towards the end of the rains. Broadcast the fertilizer in a ring about 60cm to 90cm away from the trunk to align with the feeder roots.

CROP PROTECTION

Integrated Pest Management (IPM) is crucial to protect the pods and foliage.

Major Pests: Mirids (Capsids) are the most destructive insect pests, causing shoot dieback and pod damage. Cocoa pod borers and stem borers are also common.

Control: Cultural control involves maintaining proper shade and avoiding stress on the trees. Chemical control involves spraying approved systemic or contact insecticides during peak infestation periods, which typically align with the main growing and fruiting seasons.

CHEMICAL SPRAY ON DISEASES

Fungal diseases thrive in the humid, shaded environments where cocoa grows. Chemical intervention is often mandatory to prevent total crop loss.

Black Pod Disease: Caused by *Phytophthora* fungi, this is the most devastating disease globally. It rots the pods and can spread to the stem (canker).

Chemical Control: The standard treatment is the application of copper-based fungicides (such as Bordeaux mixture, Copper oxide, or Copper sulfate) or modern systemic fungicides (like Metalaxyl).

Spraying Schedule: Spraying must be preventative. Begin applications just before the heavy rains start and continue every 3 to 4 weeks throughout the rainy season.

Safety: Always wear appropriate Personal Protective Equipment (PPE) when applying chemicals and adhere strictly to the recommended dosage to prevent chemical residues in the harvested beans.

VEGETATIVE / CLONAL PROPAGATION IN COCOA

Conventionally, cocoa is clonally propagated through budding, grafting, cutting, marcotting (air layering) and soil layering. Cutting being the most popular method.

COCOA DISEASE PEST

Black pod / Phytophthora pod rot:

Most serious disease of cocoa in West Africa, especially, Nigeria and Cameroon.

Caused by Phytophthora megakarya, during the rainy season when the relative humidity is higher than 80%.

Control of Phytophthora pod rot

The incidence of the disease is preferably prevented in the cocoa plantation through:

Frequent removal of weeds / other plants that can increase the relative humidity of the plantation.

Removal and burning of the infected cocoa pods.

Application of the copper-based fungicides to control the incidence.

The use of resistant / tolerant varieties.

Swollen shoot

A viral disease which may not appear till 6 months after cacao is infected.

Symptom appears at the shoots produced after infection.

Control measures:

Removal and burning of the infected cacao

Breeding programme has put the disease under check through the introduction of the resistant / varieties.

•Cherelle wilt

This is a physiological problem affecting only cherelles – 10cm long or less.

The cherelles suddenly wilt and die. Dead cherelles are seen hanging on cacao.

It results in 40 – 50% loss of the total pod set.

Charcoal Rot:

This is a fungal disease caused by Botryodiplodia theobromae. It is a weak pathogen. It only infects wounded, overripe or weakened cocoa pods.

•Major insect pests of cocoa

Cocoa Mirids (Capsids or Jori-jori):

Most serious insect pest of cocoa in West Africa.

The insects attack both young and mature cacao.

Mealy bugs

Mealy bugs are vectors of viral disease especially swollen shoot.

Control methods:

Chemical control method

- Biological control method

- Harvesting and post-harvest handling of cocoa

It takes 150 – 180 days between pollination and ripening in cocoa, depending on varieties.

Only mature and ripe pods are harvested, diseased and damaged pods must not be processed for markets.

Breaking / opening of cocoa pods:

Use a blunt object such as a stone or a thick piece of wood for the breaking. Extracted beans with mucilaginous pulp are collected in a clean container for fermentation.

- Preparation of Commercial Cocoa

In order to be sold as cocoa beans, the fresh cocoa removed from pods have to undergo two very important processes

- Fermentation

Heap fermentation

Basket fermentation

Sweat box fermentation

Tray fermentation

- Drying

- Drying Methods of drying.

Sun-drying

Drying autobus

Movable roof dryer.

Simple dryers

Mechanical dryers

Automated workshops.

- Grading of cocoa.

Grade 1 cocoa

Grade 2 cocoa

- Cleaning and bagging of cocoa.

- Storage of commercial cocoa.

The international standards stipulate the following conditions.

The ambient humidity must not exceed 70 %.

Periodic checking of the moisture content of each lots must be carried out.

ECONOMIC IMPORTANCE OF COCOA.

Beverages (not tea)

Foreign exchange earnings.

Black soaps

Herbs

Chocolates

The lifecycle of a cocoa tree (*Theobroma cacao*) spans several key stages: germination and nursery growth, juvenile establishment, mature fruit production, and eventual tree rejuvenation. A tree typically begins bearing pods in 3 to 4 years and reaches peak economic production by year 5, maintaining high yields for decades.

Germination and Nursery Stage (Months 1–6)

Seed Selection: Healthy, ripe cocoa pods are opened to extract fresh seeds (beans) wrapped in white pulp.

Planting: Seeds are planted directly into nursery polybags or baskets filled with rich, well-draining tropical soil.

Shade and Watering: Young seedlings require intensive shade and controlled moisture until they grow strong enough for field transfer.

Juvenile and Establishment Stage (Years 1–3)

Transplanting: Young saplings are moved to the permanent farm field, often sheltered by taller companion shade trees.

Crown Formation: The main straight trunk grows to about 1.5 meters before forming a canopy of 3 to 5 main branches.

Early Flowering: Modest white or pink flowers appear directly on the woody trunk and thick branches (a trait called cauliflory) by the third year.

Mature Production Stage (Years 5–30+)

Full Bearing: By year 5, the tree enters peak economic yield. Thousands of tiny midges pollinate the flowers, though only a small fraction develop into fruit

Pod Development: Fertilized flowers grow into large pods taking 5 to 6 months to fully ripen. Pods shift colors from green or red to bright yellow or deep purple.

Harvesting Cycles:

Ripe pods are hand-cut from trunks using sharp blades twice a year during peak seasonal crop cycles. Each pod yields about 20 to 50 beans.

Rejuvenation and Decline (Decades 30+).

Old Age: As the tree ages past 30 years, overall pod productivity declines.

Stump/Sucker Renewal: Farmers can renew old trees by allowing a strong basal shoot (sucker) to grow from the base, later cutting away the old trunk to restart the fruiting cycle.

The Yield Appreciation Curve (Biological Lifespan)

The economic life of a cocoa farm is typically modeled over a 25 to 30-year timeframe. The yield capacity per tree—and consequently per hectare—follows a distinct curve.

Years 0 to 2 (Gestation Period): The trees are establishing their root systems and canopy. There is zero yield during this phase. This period represents pure capital outflow (establishment and maintenance costs).

Years 3 to 5 (Early Bearing): The trees begin to produce pods. Yield appreciates rapidly during this phase. A well-managed farm might see yields climb from 100 kg to 400 kg of dry beans per hectare.

Years 6 to 15 (Peak Production): This is the period of maximum economic return. With optimal microclimate conditions, standard spacing (1,111 trees/hectare), and improved varieties (like Amazon hybrids), yields can stabilize between ****800 kg and 1,500 kg per hectare**** annually.

Years 16 to 25+ (Senescence): The trees begin to age, and yields naturally decline. The canopy may thin, and the trees become more susceptible to pests and diseases. Rehabilitation (such as severe pruning or grafting) or complete replanting becomes necessary.

CASHEW (*Anacardium occidentale*)



1. Description of the Cashew Plant

The cashew plant is a tropical, evergreen, perennial tree cultivated primarily for its edible kernels and the industrial liquid extracted from its shell. Characterized by a highly branched, spreading canopy and a typically short, irregular trunk, the tree can reach heights of up to 14 meters (unimproved varieties), while dwarf varieties are much shorter. It is highly valued as an industrial crop due to its three main products: the cashew kernel (highly nutritious), the cashew apple (used for juice and ethanol), and Cashew Nut Shell Liquid (CNSL) (a caustic oil used in resins, brake linings, and varnishes).

2. Origin and History

- **Origin:** The cashew tree is native to the coastal regions of northeastern Brazil.
- **History:** In the 16th century, Portuguese explorers introduced the plant to other tropical regions, starting with Goa, India, and later East Africa, primarily to control soil erosion.
- **Nigerian History:** Cashew was introduced to Nigeria around the 15th/16th century by the Portuguese. However, large-scale commercial plantations did not begin until the 1950s, pioneered in the Eastern Region (specifically Oji River) and the Western Region, eventually spreading to the Middle Belt.

3. Adaptation to Nigeria's Climatic Conditions

Cashew is a resilient crop that adapts excellently to the Nigerian climate, specifically in the savannah and derived savannah ecological zones.

- **Temperature:** Thrives in hot, tropical conditions with an ideal temperature range of 25°C to 30°C. It is sensitive to frost, which is not an issue in Nigeria.
- **Rainfall:** Requires an annual rainfall of 800 mm to 1,800 mm. It critically requires a distinct dry season (4–5 months) for prolific flowering and fruit setting. High rainfall during flowering can lead to diseases like anthracnose.

- Soils: Highly adaptable to marginal, degraded soils where other crops might fail. However, it performs best in deep, well-drained, sandy loam or lateritic soils. It cannot tolerate waterlogging.

4. Producing Areas in Nigeria

Nigeria is one of the top producers of raw cashew nuts globally. The crop is primarily cultivated in the Middle Belt, South-West, and South-East regions.

- Main Producing States: Kogi (the largest producer in Nigeria), Kwara, Oyo, Enugu, Osun, Benue, Cross River, Abia, and Imo.

5. Botany of Cashew

Cashew is a woody perennial tree characterized by a deep taproot system and extensive lateral roots, which allow it to forage for water and nutrients in poor soils and survive prolonged drought. It produces a panicle-type inflorescence (flower cluster) at the tips of new flushes of leaves.

6. Taxonomy

- Kingdom: Plantae
- Clade: Tracheophytes, Angiosperms, Eudicots, Rosids
- Order: Sapindales
- Family: Anacardiaceae (the same family as mango and pistachio)
- Genus: *Anacardium*
- Species: *A. occidentale*

7. Morphology and Anatomy

- Stem: Rough, brown bark that secretes a gum when wounded. The trunk is often crooked and branches low to the ground.
- Leaves: Simple, spirally arranged, leathery (coriaceous), obovate in shape, with prominent midribs. They are pale green when young and turn dark green as they mature.
- Flowers: The tree is andromonoecious, meaning it bears both male (staminate) and hermaphrodite (perfect) flowers on the same panicle. The flowers are small, pale green initially, turning pinkish-red upon maturity, and are highly fragrant to attract pollinators (insects/bees).

8. Structure and Form of Fruits and Seeds

What is commonly referred to as the cashew "fruit" is actually composed of two distinct parts: 1. The True Fruit (Nut): This is the kidney-shaped drupe that grows at the very end of the pseudo-fruit. It matures first. It consists of: ○ Epicarp: The leathery outer shell. ○

Mesocarp: A honeycomb structure containing the toxic, caustic oil known as Cashew Nut Shell Liquid (CNSL). ○ Endocarp: A hard inner shell enveloping the seed. ○ The Seed (Kernel): The edible part, covered by a thin, papery skin (testa). 2. The False Fruit (Pseudocarp / Cashew Apple): This is a swollen pedicel (flower stalk). It is pear-shaped, fleshy, juicy, and turns red, yellow, or orange when ripe. It is highly perishable and rich in Vitamin C.

9. Varieties and Improved Varieties

- Local Varieties: Generally characterized by tall trees, late fruiting (4–5 years), and small-sized nuts with lower yields.
- Improved Varieties: Developed and distributed in Nigeria primarily by the Cocoa Research Institute of Nigeria (CRIN). Characteristics include early bearing (18–24 months), high yield, larger nut sizes, and disease resistance. ○ Examples: Brazilian Dwarf, Jumbo Nut varieties, and specific medium-sized nut lines optimized for high kernel-outturn.

10. Cultivation Practices Nursery and Use of Poly Pots

Cashew is usually raised in a nursery before transplanting to ensure high survival rates.

- Poly Pots: Black polythene bags (about 25 cm x 15 cm) with drainage holes are used. They are filled with a topsoil-compost mixture.
- Seed Sowing: Only heavy, water-tested seeds (those that sink in water) are used. Seeds are planted 2–3 cm deep, placed flat or with the stalk-end facing upwards, to prevent the emerging root from coiling. Planting Date and Spacing
- Nursery: Seeds are planted in the nursery around January to February.
- Transplanting (Planting Date): Seedlings are moved to the permanent field at the onset of the rains (May/June) when they are 2–3 months old.
- Spacing: ○ Standard/Tall varieties: 8m x 8m to 10m x 10m. ○ Dwarf varieties: High-density planting at 4m x 4m or 5m x 5m (requires subsequent thinning as trees mature).

11. Nursery Management Practices

- Watering: Regular watering is essential, typically every 2 days during the dry season. Avoid waterlogging, which causes root rot.
- Weeding: Manual hand-pulling of weeds within the poly pots and hoeing around the nursery beds to prevent nutrient competition.
- Shading: Erect artificial shades (e.g., using palm fronds) to protect tender seedlings from intense sunlight. Shade should be gradually reduced 3–4 weeks before transplanting to "harden off" the seedlings.

12. Field Operations:

Site Selection to Transplanting

- **Site Selection:** Choose well-drained soils, avoiding marshy or waterlogged areas.
- **Land Preparation:** Slash and clear the existing vegetation. Stumping and plowing may be done for large commercial estates.
- **Lining Out and Pegging:** Use measuring tapes, ropes, and wooden pegs to mark out the exact planting spots based on the chosen spacing. This ensures straight rows for ease of mechanization and canopy management.
- **Transplanting:** Dig holes measuring 60 cm x 60 cm x 60 cm at the pegged spots. Carefully slice open the poly pot to avoid disturbing the soil ball. Place the seedling in the hole, backfill with topsoil mixed with organic matter, and firm the soil around the base.

13. Management Practices on the Plantation

- **Weed Control:** Slashing the inter-rows and ring-weeding (keeping a 1.5m radius around the tree base weed-free) are vital. Herbicides (like Glyphosate) can be used cautiously on mature plantations.
- **Mulching:** Dry leaves or cut grasses are placed around the base of the plant during the dry season to conserve soil moisture and suppress weeds. Ensure mulch does not touch the trunk to prevent termite attacks.
- **Manuring and Fertilizing:** Apply organic compost during transplanting. Inorganic fertilizers (like NPK 15:15:15 or 20:10:10) can be applied at a rate of 200–500g per tree in split doses during the rainy season. Adult trees have lower fertilizer requirements if the soil is fertile.
- **Pruning of Diseased Branches:** Lower sweeping branches, water sprouts, crisscrossing branches, and dead or diseased wood must be pruned. This creates a strong structural framework, improves aeration, and facilitates easy harvesting from the ground.

14. Diseases and Pests (With Chemical Control) Pests

- **Cashew Stem and Root Borers (*Plocaederus ferrugineus*):** The grub bores into the trunk/roots, disrupting sap flow, often killing the tree.
 - **Cashew Mosquito (*Helopeltis* spp.):** Sapsucking insects that cause lesions on young shoots, floral branches, and apples.
 - **Chemical Control:** Application of systemic and contact insecticides like Cypermethrin, Lambda-cyhalothrin, or Dimethoate (applied strictly according to agricultural guidelines).
- Diseases

- Anthracnose (*Colletotrichum gloeosporioides*): The most devastating disease. It causes water-soaked lesions, leading to leaf drop and floral dieback, severely reducing yield.
- Floral Shoot Dieback: Often a secondary infection following pest damage.
- Chemical Control: Spraying with copper-based fungicides (e.g., Copper oxychloride) or Mancozeb at the onset of flowering.

15. Harvesting, Processing, Grading, and Marketing

- Harvesting: Cashew nuts are harvested when they are fully mature. The best indicator of maturity is when the apple falls to the ground naturally. ○ Implement: Baskets (for collection), and sometimes picking sticks or sickles if the apples are required fresh. The nut is twisted off the fallen apple.
- Processing: 1. Sun-Drying: Raw nuts are dried in the sun for 2–3 days to reduce moisture to about 8–10%. 2. Roasting/Steaming: Makes the hard shell brittle and releases/extracts the caustic CNSL. 3. Shelling/Cracking: Using mechanical decorticators or manual foot-pedal crackers to extract the kernel. 4. Drying and Peeling: Kernels are dried gently to loosen the testa (papery skin), which is then peeled off.
- Grading: Kernels are graded based on color (White, Scorched), wholeness (Whole, Pieces), and size (e.g., W240, W320, where 320 means 320 nuts per pound).
- Marketing: Raw Cashew Nuts (RCN) are heavily exported from Nigeria to India and Vietnam. Processed kernels are sold in local and international markets at premium prices. CNSL is sold directly to chemical and manufacturing industries.

16. Lifecycle Cashew is a perennial crop.

- Gestation Period: Starts bearing flowers and fruits between 2 to 4 years after planting (18 months for dwarf varieties).
- Peak Production: The tree reaches its maximum yield potential between 10 and 15 years of age.
- Economic Lifespan: A well-managed cashew plantation remains economically productive for 30 to 40 years, after which yields decline, and replanting is advised.

17. Yield Potential and Profitability

- Yield Potential: Yields depend heavily on variety and management. Traditional varieties yield between 4–10 kg of raw nuts per tree per year. Improved varieties (like Brazilian Dwarf) can yield 20–30 kg per tree per year under optimal management. High-density farms can produce 1 to 2 metric tons per hectare.
- Profitability: Cashew is a highly lucrative industrial crop. The profitability stems from the high global demand for cashew kernels as a premium snack/ingredient and the industrial demand for CNSL. Although the initial capital outlay is tied up during the

gestation period, maintenance costs are relatively low once the canopy closes. Intercropping with arable crops (like groundnut or maize) during the first 3 years can provide early cash flow, making the investment highly profitable in the long term.

OIL PALM (*Elaeis guineensis*)



1. DESCRIPTION OF THE OIL PALM PLANT

The oil palm (*Elaeis guineensis*) is a perennial, unbranched, single-stemmed monocotyledonous tree. It is arguably the most efficient oil-producing crop in the world. As a major industrial crop, it yields two distinct types of vegetable oil: Crude Palm Oil (CPO), extracted from the fleshy mesocarp of the fruit, and Palm Kernel Oil (PKO), extracted from the seed (kernel). These oils are indispensable in food processing, cosmetics, biofuels, and the manufacturing of soaps and detergents.

2. ORIGIN AND HISTORY

- Origin: The crop is indigenous to the tropical rainforest region of West Africa, specifically the belt stretching from Guinea to Angola (the Gulf of Guinea).
- History:

Oil palm has been utilized by indigenous West African populations for millennia for food, medicine, and building materials. Commercial exploitation and export began during the European industrial revolution in the 19th century. Nigeria was the world's leading producer and exporter until the 1960s, a position later overtaken by Malaysia and Indonesia. Research efforts in Nigeria are spearheaded by the Nigerian Institute for Oil Palm Research (NIFOR), established near Benin City in 1939.

3. Adaptation to Nigeria's Climatic Conditions

Oil palm thrives excellently in the tropical rainforest and derived savannah ecological zones of Nigeria.

- Rainfall: Requires an annual rainfall of 1,500 mm to 3,000 mm, evenly distributed throughout the year, with a dry season lasting no more than 2–3 months.
- Temperature: Optimum temperatures range between 25°C and 33°C. It requires abundant and intense sunshine for optimal photosynthesis and fruit bunch production.
- Humidity: High relative humidity (above 75%) is essential.

- Soils: Adapts to a wide range of soils but performs best in deep, well-drained, acidic loamy soils (pH 4.0 to 6.0) that are rich in organic matter.

4. PRODUCING AREAS IN NIGERIA

Commercial production is heavily concentrated in the humid forest zones.

- Main Producing States: Edo (home to NIFOR and major commercial plantations), Delta, Ondo, Cross River, Rivers, Akwa Ibom, Imo, and Abia.

5. BOTANY OF OIL PALM

Oil palm is a monoecious crop, meaning it bears separate male and female inflorescences (flower clusters) on the same tree, though usually in alternating cycles to prevent self-pollination. Pollination is primarily entomophilous (insect-driven), largely carried out by the pollinating weevil, *Elaeidobius kamerunicus*, introduced to plantations globally to improve fruit set.

6. TAXONOMY

- Kingdom: Plantae
- Clade: Tracheophytes, Angiosperms, Monocots
- Order: Arecales
- Family: Arecaceae (Palm family)
- Genus: *Elaeis*
- Species: *E. guineensis*

7. MORPHOLOGY AND ANATOMY

- Roots: Possesses an extensive fibrous root system. Primary roots anchor the palm, while secondary and tertiary rootlets form a dense mat in the top 50 cm of the soil to absorb nutrients and water.
- Stem (Stipe): An unbranched, cylindrical trunk. In young palms, it is covered with persistent, rough leaf bases. As the tree ages (usually after 10–12 years), these leaf bases fall off, leaving a smooth stem.
- Leaves (Fronds): Large, pinnately compound leaves that can reach 7 to 8 meters in length. A mature palm maintains a crown of 40 to 50 active fronds.

8. STRUCTURE AND FORM OF FRUITS AND SEED

The oil palm fruit is a sessile drupe that grows in large, compact clusters known as Fresh Fruit Bunches (FFB).

- Exocarp: The tough, smooth outer skin, which changes color (usually from black to orange-red) as the fruit ripens.
- Mesocarp: The thick, fibrous, fleshy middle layer containing Crude Palm Oil (CPO).
- Endocarp (Shell): The hard, woody inner shell protecting the seed.
- Endosperm/Kernel (Seed): The whitish inner seed from which Palm Kernel Oil (PKO) is extracted.

9. VARIETIES AND IMPROVED VARIETIES

Classification is primarily based on the thickness of the shell (endocarp).

- Dura: Has a thick shell (2–8 mm), a large kernel, and a thin mesocarp. Yields less palm oil but more kernel oil.
- Pisifera: Shell-less, with a very thick mesocarp and a small kernel. It is female-sterile and frequently aborts its fruit, making it unsuitable for commercial planting.
- Tenera (Improved Variety): A hybrid produced by crossing Dura and Pisifera (D x P). It features a thin shell, a thick mesocarp (high CPO yield), and a distinct ring of fibers around the shell. Tenera is the standard commercial variety used globally.

10. CULTIVATION PRACTICES NURSERY AND USE OF POLY POTS

The nursery phase is critical and usually divided into a two-stage system:

- Pre-nursery: Germinated seeds are planted in small black poly pots (15 cm x 15 cm) filled with sterilized topsoil. Seedlings remain here for 4–5 months.
- Main nursery: Seedlings are transferred to larger poly pots (40 cm x 40 cm) and spaced wider apart for another 8–10 months before field planting. Planting Date and Spacing
- Planting Date: Field transplanting is done at the onset of the steady rains (usually April to June in Nigeria) to allow roots to establish before the dry season.
- Spacing: Equilateral triangular spacing of 9m x 9m x 9m is standard. This minimizes leaf overlapping and gives a planting density of about 143 palms per hectare.

11. NURSERY MANAGEMENT PRACTICES

- Watering: Young seedlings are highly susceptible to drought. Irrigation (using sprinklers or manual watering) must be done daily during dry spells.
- Weeding: Strict manual hand-weeding within poly pots is required to prevent weed competition.

- Shading: Slat houses or palm fronds are used in the pre-nursery to provide partial shade. Shade is gradually removed in the main nursery to "harden off" the seedlings for the harsh field environment.

12. SITE SELECTION, LAND PREPARATION, LINING OUT, AND PEGGING

- Site Selection: Deep, well-drained soils on flat or gently sloping land are preferred.
- Land Preparation: Involves felling of trees, burning/clearing of debris, and sometimes mechanical plowing. Sowing leguminous cover crops (e.g., *Pueraria phaseoloides* or *Mucuna bracteata*) immediately after clearing prevents soil erosion and fixes atmospheric nitrogen.
- Lining Out and Pegging: Using ropes and a surveyor's compass, the field is marked in a 9m triangular grid. Wooden pegs are driven into the ground to mark the exact planting spots.

13. TRANSPLANTING Planting holes measuring 60 cm x 60 cm x 60 cm are excavated at the pegged locations. The base of the poly pot is sliced off, the seedling is placed in the hole, and the rest of the polythene is carefully removed. The hole is backfilled with topsoil mixed with rock phosphate or organic compost, ensuring the seedling's collar is level with the soil surface.

14. MANAGEMENT PRACTICES ON THE PLANTATION

- Crop Protection: Wire mesh collars are placed around the base of newly planted seedlings to protect them from rodent attacks (e.g., cane rats).
- Weed Control: Ring weeding (keeping a 1.5–2 meter radius around the palm clear of weeds) is mandatory. Slashing or cautious herbicide application (e.g., Glyphosate) is used in the inter-rows.
- Manuring and Fertilizing: Oil palm is a heavy feeder, especially of Potassium and Nitrogen. NPK-Mg fertilizer mixtures are applied in the weeded ring. Magnesium deficiency (orange spotting on leaves) is common and corrected with Kieserite.
- Mulching: Empty Fruit Bunches (EFB) from the processing mill are spread around the base of the palms to retain moisture, suppress weeds, and recycle nutrients.
- Pruning of Diseased Branches: "Sanitation pruning" removes dead, drying, or diseased lower fronds. It improves access for harvesting and reduces breeding grounds for pests. Over-pruning must be avoided, as it reduces the photosynthetic area.

15. DISEASES AND PESTS (WITH CHEMICAL CONTROL) PESTS

- Rhinoceros Beetle (*Oryctes rhinoceros*): The adults bore into the growing point (spear leaf) of young palms, causing leaf malformation or tree death. Controlled by applying Cypermethrin or Lambda-cyhalothrin into the leaf axils.

- Leaf Miners (Coelaenomenodera elaeidis): Larvae mine into leaflets, causing desiccation. Controlled via systemic insecticides or biological control. Diseases

- Vascular Wilt (Fusarium oxysporum f. sp. elaeidis): A lethal fungal disease blocking water transport, causing fronds to snap and the tree to die. There is no effective chemical cure; management relies on planting wilt-resistant varieties.

- Ganoderma Basal Stem Rot: A fungus that decays the trunk base.

- Anthracnose / Freckle Disease: Affects seedlings in the nursery. Controlled by spraying copper-based fungicides or Mancozeb.

16. Harvesting, Processing, Grading, and Marketing

- The criterion for ripeness is when at least 1 to 5 loose fruits detach from the bunch and fall to the ground.

- Implement: A broad-bladed chisel is used for young, short palms (up to 3 meters). A Malayan knife (a curved sickle attached to a long bamboo or aluminum pole) is used for taller palms. Processing must occur within 24 hours of harvest to prevent the buildup of Free Fatty Acids (FFA).

1. Sterilization: Bunches are steamed under pressure to halt enzyme (lipase) activity and loosen fruits.

2. Stripping: Mechanical threshing separates fruits from the bunch stalk.

3. Digestion: Fruits are mashed in a heated vessel to rupture the oil cells.

4. Pressing: The mash is squeezed in a mechanical press to extract CPO, water, and impurities.

5. Clarification: The oil is separated from water and sludge through heating and centrifugation.

Grading and Marketing

- Grading: Palm oil is graded based on its Free Fatty Acid (FFA) content. Special Palm Oil (SPO) has an FFA of less than 5% (edible grade). Technical Palm Oil (TPO) has an FFA above 5% (used for soap making).

- Marketing: CPO and PKO are sold locally to refineries, food companies, and cosmetics manufacturers, or exported.

17. LIFECYCLE

- Gestation Period: Improved Tenera varieties begin to produce male and female flowers about 2 to 3 years after field planting.

- Peak Production: Reaches maximum yield potential between 8 and 15 years of age.

- **Economic Lifespan:** A commercial oil palm plantation remains highly productive for 25 to 30 years. Afterward, the trees become too tall for economical harvesting, and the plantation is felled and replanted.

18. Yield Potential and Profitability

- **Yield Potential:** Traditional, unmanaged groves yield between 2–5 tons of Fresh Fruit Bunches (FFB) per hectare per year. Well-managed plantations using improved Tenera materials can yield 15 to 25 tons of FFB/ha/year (which translates to about 3 to 5 tons of CPO per hectare).

- **PROFITABILITY:** Oil palm is one of the most profitable agricultural enterprises in the tropics. Despite the 3-year waiting period for initial returns, its 25-year productive lifespan guarantees long-term wealth generation. The crop has a zero-waste profile: oil is sold for premium prices, kernel cake is sold for livestock feed, empty bunches serve as fertilizer, and palm fronds are used for biomass energy, maximizing farm revenue.

RUBBER (*Hevea brasiliensis*)



1. DESCRIPTION OF THE RUBBER

Plant Natural rubber (*Hevea brasiliensis*) is a quick-growing, deciduous, tropical perennial tree that can reach heights of up to 30–40 meters in the wild, though cultivated trees on plantations are typically maintained at 15–25 meters. It is characterized by a straight trunk, smooth light-gray or reddish-brown bark, and a prominent canopy. Its primary industrial value lies in its latex—a milky white or yellowish fluid containing microscopic rubber particles dispersed in water—extracted via controlled tapping of the bark. Rubber latex is processed into dry natural rubber (DNR) for tires, industrial belts, footwear, medical gloves, adhesives, and sealants.

2. ORIGIN AND HISTORY

- **Origin:** *Hevea brasiliensis* is native to the Amazon Basin in South America (covering parts of Brazil, Peru, Colombia, and Ecuador).
- **History:** Indigenous South American tribes used cured rubber for balls, footwear, and waterproofing long before European contact. In 1876, English explorer Sir Henry Wickham smuggled over 70,000 rubber seeds from Brazil to the Royal Botanic Gardens, Kew, England. Seedlings grown at Kew were sent to Ceylon (Sri Lanka), Singapore, and Malaysia, laying the foundation for modern global rubber plantations.
- **Nigerian History:** Commercial rubber cultivation was introduced into Nigeria around 1894 by British colonial administrators. The Rubber Research Institute of Nigeria (RRIN), established in 1961 in Iyanomo, near Benin City, Edo State, became the hub for research and improvement of rubber production in the country.

3. ADAPTATION TO NIGERIA'S CLIMATIC CONDITIONS

Natural rubber thrives in tropical rainforest climates characterized by heat, high humidity, and abundant rainfall:

- **Rainfall:** Prefers a well-distributed annual rainfall of 2,000 mm to 4,000 mm without prolonged drought periods.
- **Temperature:** Requires high temperatures ranging between 25°C and 34°C (optimum average ~28°C). Cold temperatures or temperatures below 15°C severely retard growth.
- **Humidity:** High relative humidity (70%–80%) promotes healthy leaf development and reduces latex coagulation during tapping.
- **Soils:** Thrives best in deep, well-drained, acidic to neutral soils (pH 4.5–6.0) rich in organic matter. Clay-loam or sandy-clay loam soils in the humid forest zones are ideal.

4. PRODUCING AREAS IN NIGERIA

Rubber cultivation in Nigeria is concentrated in the southern humid forest belt:

- **Main Producing States:** Edo (the primary producer and hub of rubber research), Delta, Cross River, Akwa Ibom, Ogun, Ondo, Rivers, Abia, and Bayelsa.

5. BOTANY OF RUBBER

Hevea brasiliensis is a deep-rooted perennial woody tree. The tree features specialized laticiferous vessels (latex tubes) located within the phloem (inner bark) arranged in concentric rings. Tapping cuts into these vessels, releasing latex under pressure without damaging the vascular cambium.

6. TAXONOMY

- **Kingdom:** Plantae
- **Order:** Malpighiales
- **Family:** Euphorbiaceae (Spurge family)
- **Genus:** *Hevea*
- **Species:** *Hevea brasiliensis* (Willd. ex A.Juss.) Müll.Arg.

7. MORPHOLOGY AND ANATOMY

- **Root System:** Features a strong, deep central taproot (reaching up to 3–4 meters) complemented by an extensive network of lateral roots spread across the top 30–60 cm of soil.

- Stem/Bark: Smooth, pale grey or brownish bark. The internal anatomy of the bark consists of outer cork, hard bark, soft inner bark (rich in latex-producing laticifers), and the cambium layer.
- Leaves: Trifoliate (compound leaves with three leaflets), spirally arranged on long petioles. Leaves undergo wintering (seasonal leaf shedding during the dry season) followed by rapid flushing of new pinkish-green foliage.
- Flowers: Monoecious (male and female flowers borne on the same panicle in leaf axils). Male flowers are small and numerous, while female flowers are larger and located at the tips of main branches.

8. STRUCTURE AND FORM OF FRUITS AND SEED

- Fruit: A large, three-lobed, woody capsule containing three seeds. When mature, the dry fruit opens explosively (dehiscence) during the hot dry period, throwing seeds up to 15–20 meters away.
- Seed: Ovoid/elliptical in shape, featuring a hard, shiny, brown seed coat (testa) mottled with darker brown spots. Seeds are high in oil content (~40–50% rubber seed oil) but lose viability rapidly (recalcitrant seeds) within a few days to weeks of collection if not stored properly.

9. VARIETIES AND IMPROVED VARIETIES

Natural rubber is propagated via clonal materials (buddings) to ensure uniform yield and disease resistance.

- International Clones: RRIM 600, GT 1, PB 260, PB 217, AVROR 2037.
- Nigerian Clones (Developed by RRIN):
 - NIG 800 Series: (e.g., NIG 800, NIG 801, NIG 802, NIG 803, NIG 804) - Selected for high latex yields, rapid girth increment, wind resistance, and tolerance to leaf blight.
 - NIG 900 Series: High-yielding clone series suited for the rainforest ecology of Southern Nigeria.

10. CULTIVATION PRACTICES NURSERY AND USE OF POLY POTS

- Poly Pots: Heavy-duty black polythene bags (approx. 40 cm x 20 cm) filled with fertile topsoil mixed with compost or decomposed manure are used to raise budded stumps or polybag plants.
 - Budding: T-budding (green budding or brown budding) is conducted on 6- to 12-month-old seedling rootstocks using budwood from certified high-yielding clones.
- Planting Date
- Planting/transplanting into the field is carried out early in the rainy season (May to July) when the soil is thoroughly moist, allowing root systems to establish before dry spells.
- Spacing

11. NURSERY MANAGEMENT PRACTICES

- Weeding: Regular hand-weeding within polybags and clean hoeing between rows to prevent nutrient, light, and moisture competition.
- Shading: Slat or thatch screens (or shading net at 50% opacity) protect young seedlings and freshly budded plants from direct sun damage. Shade is gradually reduced to harden off plants prior to field establishment.
- Watering: Daily overhead watering during dry periods, ensuring adequate drainage to prevent damping-off diseases.

12. SITE SELECTION, LAND PREPARATION, LINING OUT, AND PEGGING

- Site Selection: Deep, well-drained loamy soils on flat to gently undulating terrain. Avoid waterlogged areas and shallow soils with high water tables or hardpans.
- Land Preparation: Clearing, stumping, and plowing/harrowing. Leguminous cover crops (e.g., *Mucuna mucunoides*, *Pueraria phaseoloides*) are sown to cover exposed soil, prevent erosion, and fix nitrogen.
- Lining Out and Pegging: Measuring tapes, sighting rods, and wooden pegs establish straight planting rows according to designed spacing.

13. TRANSPLANTING

- Method: Pits of 60{cm} x60{cm}x60{ cm} are dug at pegged spots and backfilled with topsoil mixed with organic matter or rock phosphate.
- Procedure: Carefully cut off the base of the polybag, place the seedling into the pit, and slice the bag vertically to remove it, fill remaining space with topsoil, and firm firmly. Ensure the bud-union stays slightly above ground level.

14. MANAGEMENT PRACTICES ON THE PLANTATION

- Crop Protection: Protecting young bark from sun scald by painting trunks with lime-wash or white latex paint up to 1 meter.
- Weed Control: Ring weeding (1.5–2 meter radius around tree bases) combined with slashing inter-rows or controlled application of systemic herbicides (e.g., Glyphosate).
- Manuring and Fertilization: Apply NPK fertilizers based on soil tests (typically NPK 15:15:15 or 12:12:17+2Mg) annually during early growth years. Phosphate fertilizers promote root development.
- Mulching: Apply dry vegetable matter or mulch around tree bases during early dry seasons to preserve soil moisture and regulate soil temperature.

- Pruning: Pruning off low side branches (up to 2.5–3 meters) to maintain a smooth, straight, unbranched trunk (virgin bark) ideal for tapping. Remove diseased or dead branches promptly.

15. DISEASES AND PESTS (WITH CHEMICAL CONTROL) PESTS

- Termites (*Coptotermes* spp.): Attack roots and inner wood, leading to tree collapse. ○ Chemical Control: Apply insecticides such as Fipronil or Chlorpyrifos at the root zone.
- Scale Insects and Mealybugs: Sap-sucking insects causing leaf distortion and soot development. ○ Chemical Control: Systemic insecticides like Imidacloprid.
- Diseases
- White Root Rot (*Rigidoporus microporus*): Devastating fungal disease causing root decay and tree death. ○ Chemical Control: Root-drenching with fungicides like Hexaconazole, Propiconazole, or Tridemorph.
- South American Leaf Blight (SALB) (*Pseudocercospora ulei*): High biosecurity risk globally; causes defoliation. ○ Chemical Control: Protective spraying with Mancozeb or triazole fungicides.
- Black Stripe / Bark Rot (*Phytophthora palmivora*): Attacks the tapping panel. ○ Chemical Control: Application of metalaxyl-based fungicides (e.g., Metalaxyl-Mancozeb) directly onto tapping cuts.

16. Harvesting, Processing, Grading, and Marketing Harvesting (Tapping)

Tapping begins when trees reach a trunk girth of 50 cm at a height of 125 cm from the bud union (typically 5 to 7 years after field planting).

- Implement: Tapping Knife (e.g., Jebong or Michie-Golledge knife), Spout, Receiving Cup (glass, plastic, or clay), Cup Hanger, and Collection Buckets.
 - Process: A thin slice of bark (1.5–2.0 mm) is shaved off diagonally at a 30° angle to the horizontal (from top-left to bottom-right) to cut laticifers. Latex flows along the cut into the spout and collects in the cup.
- Processing
1. Latex Collection: Raw latex is gathered in buckets and transported to processing centers.
 2. Coagulation: Dilute latex mixed with formic acid or acetic acid in coagulation tanks to solidify rubber into slabs/sheets.
 3. Milling: Solidified slabs are passed through grooved rollers (creping/ribbing mills) to squeeze out serum and print pattern grooves (Ribbed Smoked Sheets - RSS).
 4. Drying/Smoking: Sheets are dried in smokehouses at controlled temperature.

- Grading: Dry rubber is graded as Ribbed Smoked Sheets (RSS 1 to RSS 5) or Technically Specified Rubber (TSR / Technically Specified Rubber Nigeria - TSR-N) based on dirt content, ash content, volatile matter, and plasticity.
- Marketing: Raw crumb or smoked sheets are sold to local rubber processing plants or exported directly to tire and industrial goods manufacturers globally.

17. LIFECYCLE

- Gestation Period: 5 to 7 years from field planting to tapping readiness.
- Tapping/Economic Phase: Tapping continues productively for 25 to 30 years under proper bark management (alternating panel systems).
- End of Cycle: After 30 years, latex yields decline, and trees are felled for high-value rubberwood timber (used in furniture and wood panel manufacturing) before replanting.

18. YIELD POTENTIAL AND PROFITABILITY

- Yield Potential: ○ Unimproved/Unmanaged stands: 400 - 800kg/ha/year} of dry rubber. ○ Improved Clones (e.g., NIG series, RRIM 600) with good agronomic management: 1,500 - 2,500kg/ha/year} of dry rubber.
- **Profitability:** Rubber offers long-term financial security. While capital investment is high during the 5–7 year gestation period, once tapping begins, plantations yield steady, year-round daily or alternate-day cash flow for nearly three decades. Additional revenue from intercropping in early years, rubber seed oil extraction, and end-of-life timber sales makes commercial rubber farming highly profitable in the forest zones of Southern Nigeria.